MIL-L-13629C
7 May 1974
SUPERSEDING
MIL-L-13629B
12 September 1962

MILITARY SPECIFICATION

LACQUER: AIR-DRYING, FOR CRIMPED PARTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of lacquer for use on crimped parts.

1.2 Classification. The lacquer shall be of the following types, as specified (see 6.2):

Type I - For application to brass (oxidized).
Type II - For application to steel (phosphatized).
Type III - For application to aluminum (anodic or chemical film).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-250/8	- Aluminum Alloy 5052, Plate and Steel.
QQ-B-613	- Brass, Leaded and Non-Leaded: Flat Products (Plate, Bar, Sheet, and Strip).
QQ-S-698	- Steel, Sheet and Strip, Low-Carbon.
RR-S-366	- Sieve, Test.
TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-P-143	- Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

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Military

MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, For Aluminum and Aluminum Alloys.

STANDARDS

Federal

FED. TEST METHOD STD. No. 141	- Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.
FED. STD. No. 595	- Colors.

(Copies of specifications, and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

G 26 - Operating Light-and-Water-Exposure Apparatus (Xenon-Arc Type) for Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AIR POLLUTION CONTROL DISTRICT

Rule 66

(Application for copies should be addressed to the Air Pollution Control District, County of Los Angeles, CA 90001.)

3. REQUIREMENTS

3.1 Description. The formulation and choice of compounding material for each type lacquer (see 1.2) are optional with the supplier, provided that the finished product conforms to the requirements of this specification. Only a thinner formulated for each shipment shall be used with the particular lacquer for which it is formulated.

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3.2 Before application.

3.2.1 Unthinned lacquer.

3.2.1.1 Nonvolatile matter. The lacquer shall contain a minimum of 31.5 percent by weight of nonvolatile matter.

3.2.1.2 Volatile matter. The volatile portion of the lacquer shall conform to the requirements of Rule 66.

3.2.1.3 Weight per gallon. The lacquer shall have a minimum weight per gallon of 8.0 lb. at 73.4° F plus or minus 1° F.

3.2.1.4 Condition in container. The lacquer in a freshly opened container shall show no skinning, thickening, livering, or gelatinous masses. After standing, any separation of the ingredients of the lacquer shall be limited to that which, by manual stirring, can be reincorporated to form a homogeneous mixture.

3.2.1.5 Coarse particles. The lacquer shall contain not more than 0.1 percent by weight of particles retained on a No. 325 U.S. Standard sieve conforming to RR-S-366.

3.2.1.6 Flash point. The flash point of the unthinned lacquer shall be 35° F or greater.

3.2.2 Thinned lacquer.

3.2.2.1 Viscosity. The viscosity of the lacquer, when diluted 1:1 with thinner, shall be 26 seconds plus or minus 1.5 seconds.

3.2.2.2 Reducibility and dilution stability. The lacquer shall be completely miscible with its respective thinner in the proportion of one volume of lacquer to two volumes of the thinner. The reduced lacquer after thorough stirring shall show no separation, curdling, or precipitation.

3.2.2.3 Dry opacity. The pigmented lacquer shall have a minimum dry opacity of 0.94.

3.2.2.4 Working properties. The lacquer, when applied by spraying, shall work freely and shall show no running or sagging.

3.3 After application.

3.3.1 Drying time. The lacquer shall dry hard within 7 minutes.

3.3.2 Self-lifting property. The lacquer, applied to a priming coat of the same material which has been air dried at 80° F plus or minus 10° F for 24 hours, shall show no lifting.

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3.3.3 Appearance. The dried film shall be free from streaks, blisters, or other surface irregularities.

3.3.4 Color. Unless otherwise specified (see 6.2), the color of the pigmented lacquer film shall match color chip No. 36231 of FED. STD. No. 595.

3.3.5 Gloss. The dried film shall have a specular gloss of not more than 20.

3.3.6 Hardness. When tested as specified in 4.4.7, a lacquer film 0.5 mil plus or minus 0.1 mil thick on a sheet of plate glass, after being aged for 5 days at 75° F plus or minus 5° F, shall show a minimum hardness of 24.

3.3.7 Odor. There shall be no residual odor from a film of lacquer 48 hours after its application.

3.3.8 Water resistance. A lacquer film on a brass, steel, or aluminum panel, after being aged for 5 days at 80° F plus or minus 10° F, shall show no sign of cracking, blistering, or appreciable whitening when subjected to the action of water and, after this treatment, the film shall withstand bending without chipping, cracking, or peeling.

3.3.9 Adhesion and flexibility after accelerated aging. When tested as specified in 4.4.10, a lacquer film after being aged 30 hours at 221° F plus or minus 3.6° F shall not crack, chip, or peel

3.3.10 Accelerated weathering. When tested as specified in 4.4.11, a lacquer film shall show no change in color and no film deterioration such as cracks, crazes, or blisters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with FED. TEST METHOD STD. No. 141, Method 1031.

4.3.2 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.1 through 4.4.11 inclusive. Failure of any test shall be cause for rejection of the lot from which the sample was taken.

4.4 Test procedures.

4.4.1 Physical and chemical properties. The following tests shall be conducted in accordance with FED. TEST METHOD STD. No. 141, methods as specified in Table I. Failure of any test result to fall within the ranges specified in 3.2 or 3.3, as applicable, shall constitute failure of the applicable test.

Table I. Tests and Methods

Test		Method
Nonvolatile matter		4044
Weight per gallon		4184
Condition in container		3011
Coarse particles		4092
Viscosity		4282
Reducibility and dilution stability		4203
Dry opacity	1/	4121
Color		4250
Gloss		6101
Flashpoint		4291

1/ Use Procedure B - Doctor blade application.

4.4.2 Preparation of test panels.

4.4.2.1 Material for test panels.

4.4.2.1.1 Brass panels for Type I lacquer. Cut 3- by 5-inch panels from quarter-hard or half-hard brass conforming to QQ-B-613, Composition 268, approximately 0.015 inch thick.

4.4.2.1.2 Steel panels for Type II lacquer. Cut 3- by 5-inch panels from cold rolled sheet steel conforming to QQ-S-698, commercial quality, approximately 0.015 inch thick.

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4.4.2.1.3 Aluminum panels for Type III lacquer. Cut 3- by 5-inch panels from half-hard sheet aluminum conforming to QQ-A-250/8, temper optional, approximately 0.015 inch thick.

4.4.2.2 Treatment of test panels before application of lacquer.

4.4.2.2.1 Brass panels. Dip each panel in boiling 20-percent sodium hydroxide solution for 2 minutes, rinse thoroughly in running water for 1 minute, and then dip in concentrated hydrochloric acid (Sp. Gr. 1.19) and rinse in running water for 1 minute. Immerse the panel in a solution at 180° F plus or minus 5° F made up in the following proportions:

136 gm. sodium thiosulfate (tech.)
24 gm. lead nitrate (tech.)
17 gm. ferric nitrate (tech.)
2.4 qt. water

Treat until the surface is completely oxidized, remove from solution, rinse in water, and dry thoroughly.

4.4.2.2.2 Steel panels. The steel panels shall be phosphatized in accordance with TT-C-490, Type II. The panel may be cut from phosphatized sheet or phosphatized after cutting. The panels shall be free from foreign matter. If necessary, they shall be cleaned with an oil-free petroleum solvent or benzene and dried before coating with lacquer.

4.4.2.2.3 Aluminum panels. Prepare the aluminum panels in accordance with MIL-C-5541 or MIL-A-8625.

4.4.2.3 Application of lacquer. Spray the metal panels (oxidized brass for Type I lacquer, phosphatized steel for Type II lacquer, and anodic or chemical film aluminum for Type III lacquer) with the lacquer reduced to spraying consistency (approximately 1:1) with the thinner formulated for the lacquer under test, to produce a uniform film. Dry the test panels at 80° F plus or minus 10° F in approximately a vertical position. The dried film should be 0.5 mil plus or minus 0.1 mil thick when measured with a film-thickness gage.

4.4.3 Working properties. Visually examine the lacquer during the spraying operation specified in 4.4.2.3. Also observe the lacquer during this operation for separation, curdling, or precipitation when reducing 1:1 with thinner. Evidence of any of these defects shall constitute failure of this test.

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4.4.4 Drying time. Immediately after spraying the lacquer on the panels, as specified in 4.4.2.3, test five or more lacquered panels for dryness. Test the lacquer film at points not less than 1 inch from the top edge of the panel by touching with the finger. Tests shall be carried out in a well ventilated room maintained at a temperature of 77° F plus or minus 2° F, and relative humidity of 50 percent plus or minus 4 percent. The lacquer is considered to have dried hard when the pressure that can be exerted between the thumb and finger does not mar the film nor leave a mark which remains noticeable after the spot is lightly polished. If rapid, light rubbing breaks the surface, the sample is considered not to have dried hard. Failure to dry hard within 7 minutes shall constitute failure of this test.

4.4.5 Self lifting. Spray a second coat of the lacquer on two of the panels that have been previously sprayed and dried for 24 hours at 80° F plus or minus 10° F and examine the panels for self-lifting. Evidence of lifting shall constitute failure of this test.

4.4.6 Surface appearance. Examine visually all the lacquered panels for evidence of streaks, blisters, and other irregularities of the surface. Evidence of any of these defects shall constitute failure of this test.

4.4.7 Hardness. Spray one coat of properly thinned lacquer on a clean piece of plate glass. Dry this specimen at room temperature in approximately a vertical position. The dried film shall be 0.5 mil plus or minus 0.1 mil thick. Age for 5 days and determine the hardness by means of a Sward Hardness Rocker at 75° F plus or minus 5° F. Before obtaining hardness readings, calibrate the instrument to give a reading of 100 on a clean glass plate. Make all readings during the calibration and on the test films with the plate in a level position (see 6.3). A reading of less than 24 shall constitute failure of this test.

4.4.8 Odor. Examine the lacquered panels 48 hours after spraying for evidence of residual odor from the films. Detection of an odor shall constitute failure of this test.

4.4.9 Water resistance. Age three of the lacquered panels for 5 days at a temperature of 80° F plus or minus 10° F and immerse the panels in distilled water at room temperature for 24 hours. Remove the panels from the water at the end of the period, allow to stand for 3 hours, and examine visually for signs of checking, blistering, or appreciable whitening. After the visual examination, bend each of the lacquered panels, with the lacquered side up, through an angle of 180 degrees over a 1/2-inch metal rod, held firmly by suitable supports. If the lacquer chips or peels off any of the panels, spray five additional panels with lacquer and repeat the test. Evidence of any chipping or peeling of the lacquer from the second set of panels shall constitute failure of this test.

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4.4.10 Adhesion and flexibility after accelerated aging. Age three of the lacquered panels in an oven at 221° F plus or minus 3.6° F for 30 hours. Remove from oven, cool for 30 minutes, and bend the panels, with the lacquered sides up, through an angle of 180 degrees over a 1/8-inch metal rod held firmly by suitable supports. If the lacquer film shows any defects in adhesion and flexibility such as cracking, chipping, or peeling, repeat the test of five additional panels. Evidence of any cracking, chipping, or peeling of the lacquer from the second set of panels shall constitute failure of this test.

4.4.11 Accelerated weathering. Prepare duplicate test panels as specified in 4.4.2.2 and 4.4.2.3, as applicable. Expose one test panel for 300 hours of accelerated weathering in accordance with FED. TEST METHOD STD. No. 141, Method 6151 or ASTM Method G 26, Type BH. Compare the exposed test panel with the test panel similarly prepared but not exposed in the accelerated weathering apparatus. Evidence of any change in color and film deterioration such as cracks, crazes, or blisters shall constitute failure of this test.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined and tested to determine compliance with the quality assurance provisions of TT-P-143.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The lacquer shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be A, B, or C, and the level of packing shall be A, B, or C as specified (see 6.2). The lacquer shall be furnished in the size of container as specified (see 6.2).

6. NOTES

6.1 Intended use. The lacquer covered by this specification is intended for use as a protective coating for crimped parts.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of lacquer required (see 1.2).
(c) Color required, if other than as specified in 3.3.4.
(d) Level of packaging and level of packing required (see 5.1).
(e) Size of container required (see 5.1).

6.3 Sward Hardness Rocker. Detailed instructions on the use of Sward Hardness Rocker are available in Scientific Section Circular No. 510 of the National Coatings and Paint Association, Washington, DC 20005.

Custodians:
Army - ME
Air Force - 84

Review activities:
DSA - IS

Preparing activity:
Army - ME

Project No. 8010-0717

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